

Use the following context to answer questions 1-5.

The length of a machine part should be 5.25 inches (in). The manufacturing tolerance is within 0.03 in.

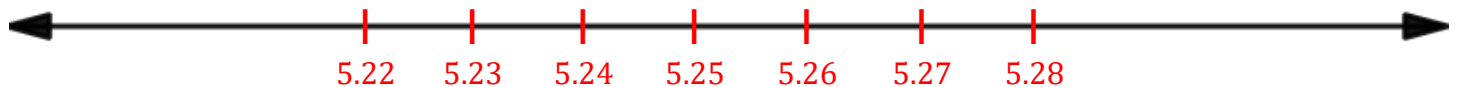
1. What is the minimum value the machine part could be?

$$5.25 - 0.03 = 5.22$$

2. What is the maximum value the machine part could be?

$$5.25 + 0.03 = 5.28$$

3. Sketch a number line that shows the minimum value, maximum value, and 5.25.



4. Write an absolute value equation that could be used to determine the maximum and minimum value of m , the length of the machine part, algebraically.

$$|m - 5.25| = 0.03$$

5. Show that the solutions to the equation match the maximum and minimum values determined in questions 1 and 2.

$$\begin{array}{r} m - 5.25 = 0.03 \\ +5.25 \quad +5.25 \end{array}$$

$$m = 5.28$$

$$\begin{array}{r} m - 5.25 = -0.03 \\ +5.25 \quad +5.25 \end{array}$$

$$m = 5.22$$

For questions 6-10, solve each absolute value equation.

6. $0 = 3|x + 2| - 15$

$$\frac{3|x+2|}{3} = \frac{15}{3}$$

$$|x + 2| = 5$$

$$\begin{array}{cc} x + 2 = 5 & x + 2 = -5 \\ -2 & -2 \end{array}$$

$$x = 3 \quad x = -7$$

7. $|-4x + 2| - 9 = 0$
 $\quad \quad \quad +9 \quad +9$

$$|-4x + 2| = 9$$

$$\begin{array}{cc} -4x + 2 = 9 & -4x + 2 = -9 \\ -2 & -2 \end{array}$$

$$\begin{array}{cc} \frac{-4x}{-4} = \frac{7}{-4} & \frac{-4x}{-4} = \frac{-11}{-4} \end{array}$$

$$x = -1.75 \quad x = 2.75$$

8. $|x + 5| + 9 = 3$
 $\quad \quad \quad -9 \quad -9$

$$|x + 5| = -6$$

No Solution

9. $|3x + 4| - 6 = 25$
 $\quad \quad \quad +6 \quad +6$

$$|3x + 4| = 31$$

$$\begin{array}{cc} 3x + 4 = 31 & 3x + 4 = -31 \\ -4 & -4 \end{array}$$

$$\begin{array}{cc} \frac{3x}{3} = \frac{27}{3} & \frac{3x}{3} = \frac{-35}{3} \end{array}$$

$$x = 9 \quad x = -11\frac{2}{3}$$

10. $-|x - 3| + 7 = 0$
 $\quad \quad \quad -7 \quad -7$

$$\frac{-|x-3|}{-1} = \frac{-7}{-1}$$

$$|x - 3| = 7$$

$$\begin{array}{cc} x - 3 = 7 & x - 3 = -7 \\ +3 & +3 \end{array}$$

$$x = 10 \quad x = -4$$